

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -1(Adv-2022-P1-Model)** **Date: 12-04-2026**

Time: 09.00Am to 12.00Pm **GTA-23** **Max. Marks: 180**

12-04-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Adv(2022-P1)_GTA-23_Syllabus,

PHYSICS : TOTAL FIRST YEAR SYLLABUS

CHEMISTRY : TOTAL FIRST YEAR SYLLABUS

MATHEMATICS : TOTAL FIRST YEAR SYLLABUS

Rankers Academy JEE
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Name of the Student: _____ **H.T. NO:** | | | | | | | |

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JEE-ADVANCED-2022-P1-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 8)	Questions with Numerical Value Answer Type	+3	0	8	24
Sec – II(Q.N : 9 – 14)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – III(Q.N : 15 – 18)	Matching Type	+3	-1	4	12
Total				**18**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 26)	Questions with Numerical Value Answer Type	+3	0	8	24
Sec – II(Q.N : 27 – 32)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – III(Q.N : 33 – 36)	Matching Type	+3	-1	4	12
Total				**18**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 37 – 44	Questions with Numerical Value Answer Type	+3	0	8	24
Sec – II(Q.N : 45 – 50)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – III(Q.N : 51 – 54)	Matching Type	+3	-1	4	12
Total				**18**	**60**

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MATHEMATICS

Max Marks: 60

SECTION – I
(NUMERICAL VALUE TYPE)

This section contains **EIGHT (08)** questions.

- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will bee evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the correct numerical value is entered ;
Partial Mark: 0 In all other cases.

1. Let $f:R\to R$ be a real valued function such that $f(x)=\left|x^2-1\right|+\left|x^2-3x+2\right|-\left[x^3+x\right]$, where [.] denotes GIF. Let ' m ' be, number of points of local minima of the function $f(x)$ and M be the number of points of local maxima of the function $f(x)$ in the interval $(0,2)$ then $m-M$ is equal to ____

2. Let $P(x)=x^4+ax^3+bx^2+cx+d=0$, has four integer roots, where **a**,**b**,**c**, and **d** are integers. The sums of the pairs of roots of $P(x)=0$ are given by 1,2,5, **6**,**9** and 10 then the value of $\left|\frac{P(0)}{7}\right|=$.....

3. Given $a,b\in\{1,2,3\}$ & $x\in\left[\frac{\pi}{2},\frac{3\pi}{2}\right]$ and number of solutions for x of the equation $2a\sin^2 x+\cos x=b\sin x$ is 1 , then number of such ordered pairs (a,b) is.....

4. The value of $8\left(\frac{1}{2}+\cos\frac{\pi}{20}\right)\left(\frac{1}{2}+\cos\frac{3\pi}{20}\right)\left(\frac{1}{2}+\cos\frac{9\pi}{20}\right)\left(\frac{1}{2}+\cos\frac{27\pi}{20}\right)$ is equal to.....

5. If $\frac{3}{1.2.4}+\frac{4}{2.3.5}+\frac{5}{3.4.6}+\cdots+\frac{n+2}{n(n+1)(n+3)}=\frac{1}{6}\left[\frac{a}{6}-\frac{4}{n+1}-\frac{1}{n+2}-\frac{b}{n+3}\right]., a,b\in I^+$

Then the value of $\frac{a+b}{12}=$.....

6. Number of roots, sum of the roots of the equation $f(x)=f\left(\frac{x+8}{x-1}\right)$, are m, n respectively. where $f(x)=x^2-12x+3$, if it is known that $f(x)$ is defined in the interval $[-5,5]$. Then the value of m + n is equal to.....

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7. Let $A=\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ where a, b, c, d are positive integers arranged in ascending order, and exactly two of a,b, c, d are prime numbers and also pair wise coprime and satisfies $A^2-16A-17I=0$ (where I is an Identity matrix of order 2×2). If $\mathrm{B}=\begin{bmatrix} \mathrm{b} & \mathrm{a} \\ \mathrm{c} & \mathrm{d} \end{bmatrix}$, then $\mathrm{Det}(B)$ equals……

8. STATEMENT 1 :Let OABC be a regular tetrahedron of side length unity, where O is origin. If P is a point at a unit distance from origin such that $\overrightarrow{OP}$ is equally inclined to $\overrightarrow{OA},\overrightarrow{OB}$ and $\overrightarrow{OC}$ at an angle α, then $p_1=\cos^2\alpha$.

STATEMENT 2 : $\mathrm{A,B,C,D}$ are four points in the space and satisfy $\mathrm{AB}=3,\mathrm{BC}=7$, $\mathrm{CD}=11$ and $\mathrm{DA}=9$, also if angle between AC and BD is θ then $p_2=\cos\theta$.

Considering above two statements, the value of $[3p_1+2p_2]$ is, (where [.] represent Greatest Integer Function)…….

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

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Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

9. Let $f:R\to R$ be a differentiable function such that $f(x+y)-f(x-y)=4xy+10y \quad \forall x,y\in\mathbb{R}$ and $f(1)=2$ then:

A) Minimum value of $f(x)$ is $\frac{-41}{4}$

B) Minimum value of $f(f(x))$ is $-\frac{41}{4}$

C) The equation $|f(|x|)|=K$ has exactly two solutions when $K\in(0,4)$

D) The equation $|f(|x|)|=4$ has exactly 3 solutions

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10. If $\begin{vmatrix} \int_0^{x^2} f(\sqrt{t})dt & f(x) & \int_0^x e^{\frac{t^2}{2}+\frac{t}{6}}dt \\ 1 & 2 & 3 \\ 4 & 5 & 6 \end{vmatrix} = \int_0^x f(t)dt \quad \forall x \in R^+ \cup \{0\}$ then choose the correct statements

A) $\lim\limits_{x\to 6} \dfrac{f(x)-3e^{19}}{x-6} = 19e^{19}$

B) Number of points where the function $y = |f(x)-a|$ is non-differentiable for $a>0$ is always equal to 1

C) Number of points where the function $y = |f(|x|)-a|$ is non-differentiable for $a>0$ is always equal to 3

D) The function $g: R^+ \to R \quad g(x) = \dfrac{f(x)}{xe^x}$ has a local minima at $x = \dfrac{5}{6}$.

11. Let **x** be a positive number, A sequence $\{x_n\}$ of real numbers is defined as follows:

$x_1 = \frac{1}{2}\left(x+\frac{5}{x}\right), x_2 = \frac{1}{2}\left(x_1+\frac{5}{x_1}\right), \ldots$, and in general, $x_{n+1} = \frac{1}{2}\left(x_n+\frac{5}{x_n}\right)$ for all $n \geq 1$. Then which of the following is / are true

A) for all $n \geq 1, \dfrac{x_n-\sqrt{5}}{x_n+\sqrt{5}} = \left(\dfrac{x-\sqrt{5}}{x+\sqrt{5}}\right)^{2^n}$ **B)** $\lim\limits_{n\to\infty} x_n = \sqrt{5}.$

C) for all $n \geq 1, \dfrac{x_n-\sqrt{5}}{x_n+\sqrt{5}} = \left(\dfrac{x-\sqrt{5}}{x+\sqrt{5}}\right)^{2^{\frac{n}{2}}}$ **D)** $\lim\limits_{n\to\infty} x_n = \dfrac{1}{\sqrt{5}}.$

12. If the equation $x^3 \sin q - (\sin q + 2)x^2 + 6x - 4 = 0, \; q \in \left(\dfrac{-\pi}{2}, \dfrac{\pi}{2}\right)$ has 3 positive roots and let

$$f(\sin q) = \frac{9\sin^2 q - 4\sin q + 3}{(1-\cos q)(2\cos q - 6\sin q - 3\sin 2q + 2)}.$$

Then which of the following is are true

A) $f(t)$ attains its minimum value in its domain

B) $f(t)$ attains its maximum value in its domain

C) $f(t)$ attains at least one local minimum in its domain

D) $f(t)$ attains at least one local maximum in its domain

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13. Let $P=\left[a_{ij}\right]_{3\times 3}$ matrix with $tr(P)=2$ such that exactly four elements of P are 1 and the rest are zero and there exists unique column matrix $M=\begin{pmatrix}x\\y\\z\end{pmatrix}$ such that $PM=0$

Then which of the following is (are) true ?

A) Number of such matrices P, is 6. **B)** $\det(P)$ has two possible values.

C) $\det(adj(P))$ can be –1 **D)** $\det(adj(P))$ can be 1

14. $ABP'CC'DB'P$ is a parallelogram, with $BB'\|CC'\|AD\|P'P$ and $AB\|CP'\|C'P\|DB'$. AB, AC, AD are adjacent edges of a parallelepiped. The vector along the diagonal AP is given by $\vec{a}$ so that $|\vec{a}|=3$. The vector areas of the faces containing A, B, C and A, B, D are respectively given by $\overrightarrow{AB}\times\overrightarrow{AC}=\vec{b};\overrightarrow{AB}\times\overrightarrow{AD}=\vec{c}$. The projections of the edges AB and AC on the diagonal AP are both equal to 1 . Then which of the following can be correct

A) $\overrightarrow{AC}=\frac{\vec{a}}{3}-\frac{1}{9}(\vec{a}\times(2\vec{b}-\vec{c}))$ **B)** $\overrightarrow{CD}=\frac{1}{3}(\vec{a}\times(\vec{b}-\vec{c}))$

C) $\overrightarrow{AB}=\vec{a}+\frac{1}{3}(\vec{a}\times(2\vec{b}+\vec{c}))$ **D)** $\overrightarrow{AD}=\frac{\vec{a}}{3}+\frac{1}{9}(\vec{a}\times(\vec{b}-2\vec{c}))$

SECTION – III
(MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosed (i.e. the question is unanswered);
Negative Marks: –1 In all other cases.

15. Let $f(x)=\cos^{-1}\left(\frac{a\cos x+b}{a+b\cos x}\right)-2\tan^{-1}\left(\sqrt{\frac{a-b}{a+b}}\tan\frac{x}{2}\right)(0<b\le a),x\ge 0.$

$$g(x)=2\tan^{-1}x+\sin^{-1}\left(\frac{2x}{1+x^2}\right),\ x\ge 1,\ h(x)=\begin{cases}f(x),x\in[0,1)\\g(x),x\in[1,\infty)\end{cases}$$

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	List 1		List 2
P	$f\left(\frac{\pi}{2}\right) =$ ____	1	0
Q	$\left[h\left(\frac{3}{2}\right)\right] =$ _____ $([.]G.I.F)$	2	1
R	$\frac{g(1)+g\left(\frac{3}{2}\right)}{g(2)} =$ ____	3	2
S	Number of integers in the range of $h(x)$ is	4	3
		5	4

A) P – 1, Q – 4, R – 3, S – 1 **B)** P – 1, Q – 2, R – 3, S – 4

C) P – 1, Q – 3, R – 2, S – 5 **D)** P – 1, Q – 4, R – 5, S – 3

16. Match List – I and List – II

	List-I		List-II
P	Number of solutions of $\cos 3x\cos 6x = \cos 4x\cos 7x, x \in \left[0, \frac{\pi}{2}\right)$	1	1
Q	Number of solutions of $\sin 2x\sin 6x = \cos x\cos 3x, x \in \left[0, \frac{\pi}{2}\right]$	2	2
R	Number of solutions of $\cos 3x\cos 7x = \cos 2x\cos 8x, x \in \left[0, \frac{\pi}{2}\right]$	3	3
S	Number of solutions of $\sin 5x\cos 3x = \sin 6x\sin 2x, x \in \left[0, \frac{\pi}{2}\right)$	4	4
		5	5

A) P – 1, Q – 4, R – 5, S – 3 **B)** P – 5, Q – 4, R – 3, S – 2

C) P – 4, Q – 5, R – 2, S – 3 **D)** P – 1, Q – 5, R – 3, S – 2

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17. Let (x_1, y_1, z_1) and (x_2, y_2, z_2) are 2 set of solution satisfying the following system of equations

$\log_{10}(2xy) = 4 + (\log_{10} x - 1)(\log_{10} y - 2)$

$\log_{10}(2xy) = 4 + (\log_{10} y - 2)(\log_{10} z - 1)$

$\log_{10}(zx) = 2 + (\log_{10} z - 1)(\log_{10} x - 1)$ such that $x_1 > x_2$,

Then match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
P)	$\frac{y_1}{x_1}$	1)	2
Q)	$\frac{z_1}{x_2}$	2)	100
R)	$\frac{z_1 x_2}{z_2}$	3)	1000
S)	$\frac{y_2 + z_1}{x_2}$	4)	150

A) P-1; Q-2; R-4; S-3 **B)** P-1; Q-2; R-3; S-4

C) P-2; Q-1; R-4; S-3 **D)** P-2; Q-1; R-3; S-4

18. Let $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ are position vectors of 4 points A, B, C, D such that

$|\vec{a}| = 1, |\vec{b}| = 2, |\vec{c}| = 3, |\vec{d}| = 4$ and $(\vec{a} \times \vec{b}).(\vec{c} \times \vec{d}) = 24$, $\vec{a}.\vec{c} = \frac{3}{2}, \vec{b}.\vec{c} > 0$, then

Area of $\Delta ABC = \frac{1}{4}(2\alpha + \beta\sqrt{3})$ and area of $\Delta BCD = \frac{1}{2}(m - n\sqrt{3})$, where $\alpha, \beta, m, n \in N$

Match List-I with List-II and select the correct answer using the code given below the list.

	List-I		List-II
P)	α	1)	1
Q)	β	2)	3
R)	m	3)	4
S)	n	4)	9

A) P-1; Q-2; R-3; S-4 **B)** P-2; Q-3; R-4; S-3

C) P-4; Q-3; R-1; S-2 **D)** P-1; Q-2; R-4; S-3

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PHYSICS **Max Marks: 60**

SECTION – I
(NUMERICAL VALUE TYPE)

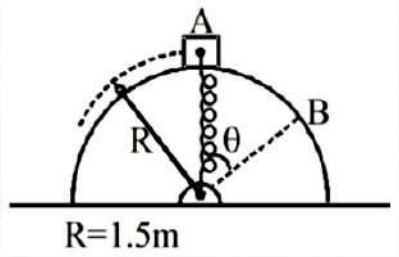
This section contains **EIGHT (08)** questions.

- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will bee evaluated according to the following marking scheme :

Full Marks: +3 **ONLY** if the correct numerical value is entered ;
Partial Mark: 0 In all other cases.

19. An open organ pipe containing air resonates in fundamental mode due to a tuning fork. The measured values of length l (in cm) of the pipe and radius r (in cm) of the pipe $l = 94 \pm 0.1, r = 5 \pm 0.05$. The velocity of the sound in air is accurately known. The maximum percentage error in the measurement of the frequency of that tuning fork by this experiment is given by $\alpha^2\%$. Find the value of 10α.

20. A small 2 kg block is gently pushed from rest at A and it slides down along the fixed circular surface as shown in figure.



If the attached spring has a force constant $k = 20\,N/m$. What is unstretched length of spring (in m) so that it does not allow the block to leave the surface until angle with the vertical is $\theta = 60^\circ$? $\left(g = 10\,m/s^2\right)$

21. A satellite of mass m is initially in a circular orbit C_1, of radius R around the Earth. In this problem, we discuss how to transfer the satellite from this initial condition to a new circular orbit C_2 of radius $2R$. To perform the desired orbital transfer, we have two impulsive velocity increments capable of producing a change $\Delta\vec{v}$ during a time interval much shorter than the satellite's orbital period. The first impulse causes the satellite to leave its original circular orbit and move along an elliptical orbit E. Next, at a certain instant, a second impulse is applied in order to transfer the satellite from the elliptical transfer orbit to the desired new circular orbit C_2. Assume that the impulses always occur in the tangential direction to the trajectories. The impulse for the first maneuver is Δv_1 and the impulse for the second maneuver is Δv_2. The value of $\left(\frac{\Delta v_1}{\Delta v_2}\right)$ is _____.

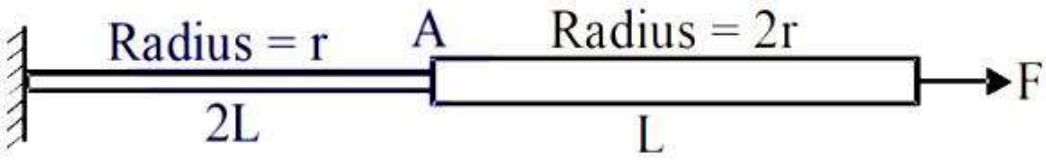
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22. The volume flow rate $\left(\text{in } m^3 s^{-1}\right)$ of a liquid through pipe having diameter d is related to viscosity of water 'η' (unit Pascal. s) and the pressure gradient along the pipe $\frac{dP}{dx}$ [pressure gradient $\frac{dP}{dx}$ is rate of change of pressure per unit length along the pipe], by a formula of the form $Q = K\eta^a d^b \left(\frac{dP}{dx}\right)^c$. Where K is a dimensionless constant. Find $a+b+c$.

23. A thermometer has a spherical bulb of volume 1 cm^3 of mercury. A long cylindrical capillary tube is connected to spherical bulb. Volumetric coefficient of expansion of mercury is $1.8\times10^{-4}K^{-1}$; cross – section area of capillary is $1.8\times10^{-4}cm^2$. Ignoring expansion of glass, how far apart (in cm) on the stem are marks indicating $1K$ temperature change?



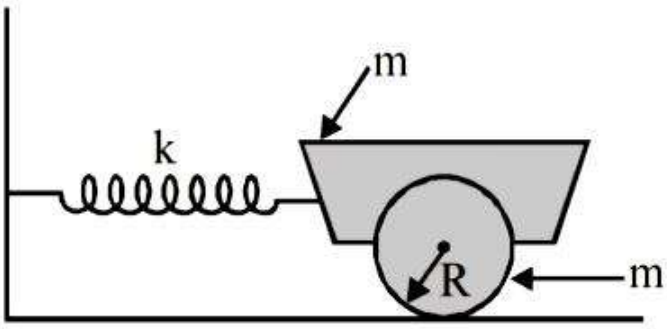
24. Two steel wires of radii r and $2r$ are connected together end to end and tied to a wall as shown. The force stretches the combination by $\frac{27}{4}mm$. How far does the junction point A move (in mm)?



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25. A string will break apart if it is placed under too much tensile stress. One type of steel has density. $\rho_{steel} = 10^4\, kg/m^3$ and breaking stress $\sigma = 9\times10^8\, N/m^2$. We make a guitar string from (4π) gram of this type of steel. It should be able to withstand $(900\pi)N$ without breaking. What is highest possible fundamental frequency (in Hz) of standing waves on the string if the entire length of the string vibrates?

26. A cart consists of a body of mass m and two wheels, each of mass m and radius R. The cart is attached to a spring of constant k. The other end of the spring is fixed to a wall as shown in figure. If time period of oscillation is $\beta\times2\pi\sqrt{\frac{m}{k}}$ then find the value of β.



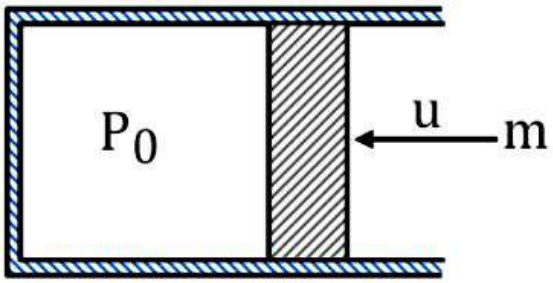
SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
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Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

27. A horizontal piston cylinder arrangement encloses a gas at a pressure P_0. The atmospheric pressure is also P_0. Now a particle of mass m strikes the piston of identical mass m with a horizontal speed u and sticks to it. Subsequently the piston moves without friction and compresses the gas. The process of compression can be assumed to be adiabatic. In this process

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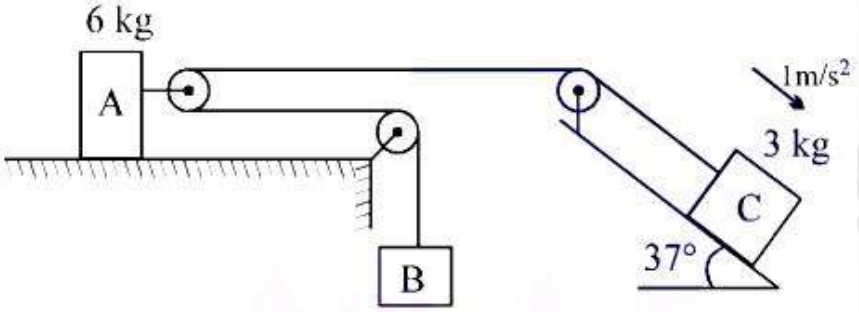
A) When the piston comes to rest at maximum compression, the internal energy of gas would have increased by $\frac{1}{4}mu^2$.

B) During the time interval starting from the mass striking the piston to the maximum compression of the enclosed gas, the work done by the gas is $-\frac{1}{4}mu^2$.

C) When the piston comes to rest after compressing the gas, the internal energy of the gas would have increased by the amount more than $\frac{1}{4}mu^2$.

D) During the compression of the gas till the time when piston comes to rest, work done by the gas is less than $-\frac{1}{4}mu^2$.

28. Three blocks A, B and C are arranged as shown. Pulleys and strings are ideal. All surfaces are frictionless. If block C is observed moving down along the incline at $1m/s^2$. Then (system is released from rest). $\left(g = 10m/s^2\right)$



A) The mass of block B is 15kg.

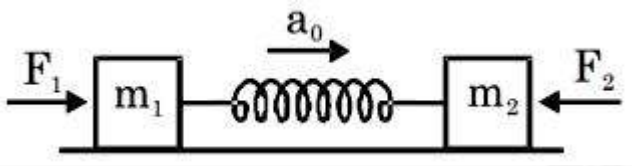
B) Tension in string is $15N$.

C) Acceleration of A and B are $5m/s^2$ and $9m/s^2$ respectively.

D) Acceleration of A and B are $9m/s^2$ and $5m/s^2$ respectively.

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29. Two blocks m_1 and m_2 are connected with a compressed spring and placed on a smooth horizontal surface as shown in figure. Force constant of spring is k. Under the influence of forces F_1 and F_2, at an instant blocks move with common acceleration a_0. At that instant force F_2 is suddenly withdrawn. Mark correct option.



A) Instantaneous acceleration of m_1 is $a_0 - \frac{F_1}{m_1}$

B) Instantaneous acceleration of m_2 is $a_2 = a_0 + \frac{F_2}{m_2}$

C) Instantaneous acceleration of m_1 is $a_1 = a_0$

D) Spring force is $F_{spring} = m_2 a_0 + F_2$

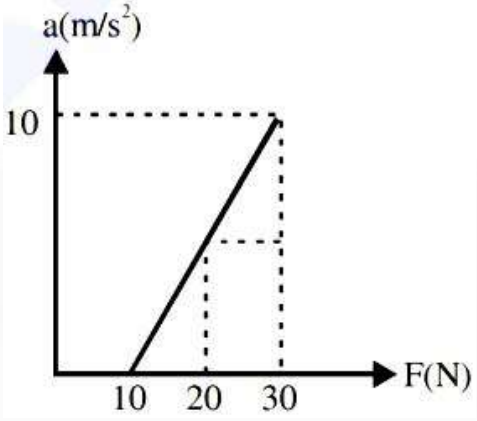
30. River is flowing with a velocity $\vec{v}_R = 4\hat{i} ms^{-1}$. A boat is moving with a velocity $\left(-2\hat{i} + 4\hat{j}\right) ms^{-1}$ relative to river and width of river is 100m along y direction then –

A) The time of crossing is 25 sec. **B)** Absolute velocity of boat is $2\sqrt{5} ms^{-1}$

C) Drift of boat is 50m. **D)** The boat can never cross the river.

31. A block placed on a rough horizontal surface is pushed with a force F acting horizontally on the block. The magnitude of F is increased and acceleration produced is plotted in the graph shown. $\left(g = 10 m/s^2\right)$

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A) Mass of the block is 2 kg,

B) Coefficient of friction between block and surface is 0.5.

C) Limiting friction between block and surface is $10N$.

D) When $F = 8N$, friction between block and surface is $10N$.

32. Mark the CORRECT statement(s):

A) When two sound waves of different frequency f_1 and f_2 are superimposed at a point, the variation of intensity takes place with frequency $|f_1 - f_2|$.

B) When an observer is moving, relative speed of the sound waves with respect to the observer is changed, while due to motion of source, the sound wavelength of the wave changes.

C) When a high pressure compression pulse reaches water surface in a resonance column, it reflects as a low pressure rare fraction pulse due to a phase change of π.

D) When a tuning fork is struck by a hammer a transverse wave sets in the prong while a longitudinal wave sets in the stem (handle) of tuning fork.

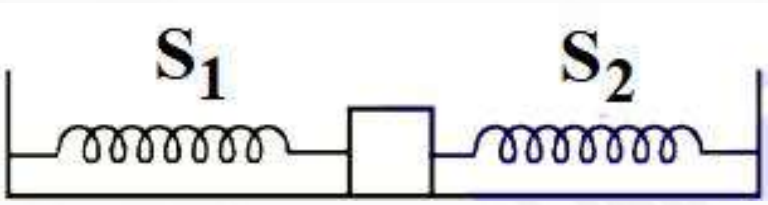
SECTION – III
(MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosed (i.e. the question is unanswered);
Negative Marks: –1 In all other cases.

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33. The figure shows a block B of mass 2 kg kept on a smooth horizontal floor in equilibrium with two identical springs of S_1 & S_2 force constant $k = 100\,N/m$ attached to it and to fixed supports as shown. The block is then displaced horizontally from this position by amount x and released. Match the initial equilibrium conditions & subsequent values of x in Column–I with corresponding acceleration of the block when released in Column–II.



Column – I		Column – II	
A	S_1 and S_2 are relaxed and $x = 2cm$	P	$6\,m/s^2$
B	S_1 and S_2 are stretched by 1 cm and $x = 2cm$	Q	$4\,m/s^2$
C	S_1 and S_2 are compressed by 2 cm and $x = 4cm$	R	$3\,m/s^2$
D	S_1 and S_2 are compressed by 4 cm and $x = 1cm$	S	$2\,m/s^2$
		T	$1\,m/s^2$

The correct match is

A) A –S, B –S, C – Q, D – T

B) A – S, B – P, C – T, D – Q

C) A – P, B –S, C –R, D – Q

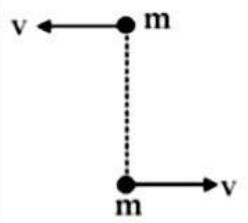
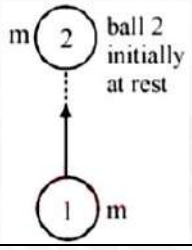
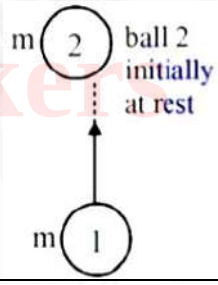
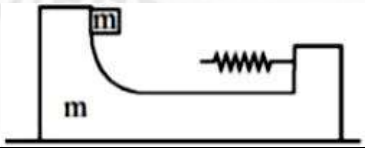
D) A – T, B – R, C – S, D – P

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34. Column – I describes the system and initial motion of its particles and Column – II gives characteristics associated with subsequent motion.

	Column – I		Column – II
A	Two identical particles under influence of mutual gravitational force only. Particles are given initial velocity perpendicular to line joining as shown. 	P	Centre of mass of system does not accelerate
B	Two identical balls on a smooth table. Elastic head on collision takes place. 	Q	In centre of mass frame linear momentum of system is zero.
C	Two identical balls on a smooth table. Oblique elastic collision takes place. 	R	Linear momentum of system is conserved.
D	Block is released on a wedge a shown in figure. All the surfaces are smooth. 	S	Mechanical energy of system is conserved.
		T	Total kinetic energy is constant

The correct match is

A) A – PQ, B – QT, C – PT, D – P **B)** A – Q, B – PQT, C – PT, D – P

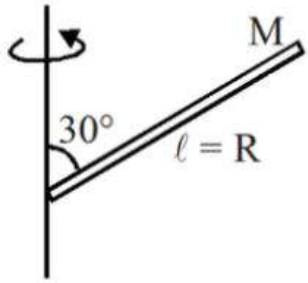
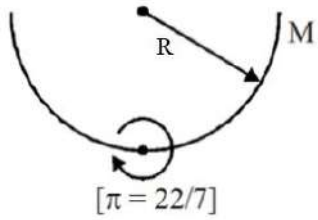
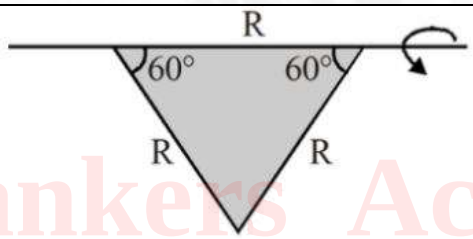
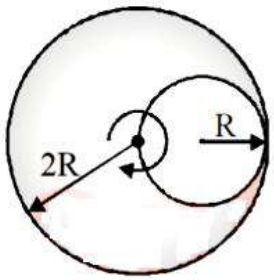
C) A – R, B – RT, C – T, D – R **D)** A – S, B – PQS, C – PT, D – PR

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35. Mark the matching options for objects shown in respective figures of Column –I with correct moment of inertia given in Column-II

Column – I		Column – II	
A	 Uniform rod	P	$\frac{8MR^2}{11}$
B	 Uniform Semicircular Ring. Axis is perpendicular to plane of ring.	Q	$\frac{MR^2}{12}$
C	 Uniform Triangular Plate of mass M	R	$\frac{13MR^2}{8}$
D	 Uniform disk of initial mass M from which circular portion of radius R is then removed. M.I. of remaining mass about axis which is perpendicular to plane of plate	S	$\frac{MR^2}{8}$

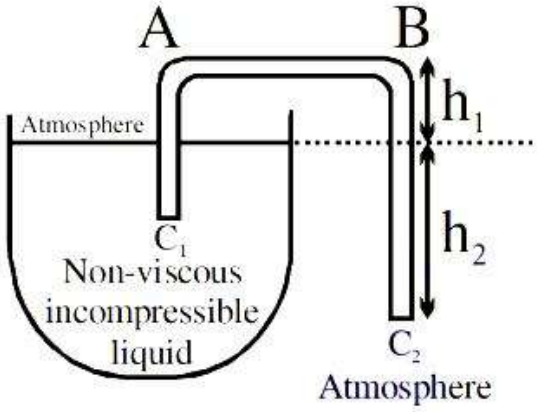
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The correct match is

A) A – Q, B – S, C – S, D – R **B)** A – R, B – P, C – Q, D – S

C) A – S, B – R, C – P, D – Q **D)** A – Q, B – P, C – S, D – R

36. An evacuated thin tube has two tight caps C_1 and C_2 at its ends. The tube is adjusted as shown in the figure below. Atmospheric pressure = P and density of the liquid = d, Neglect surface tension of liquid. Match the entities of columns.



Column – I		Column – II	
A	$h_1 > \frac{P}{dg}, h_2 = 0$	P	If C_2 is opened first and then C_1 is opened, liquid level in the left arm of the tube will be on level with liquid in the pot.
B	$h_1 > \frac{P}{dg}, h_2 > 0$	Q	If C_1 is opened with C_2 is closed, liquid will rise in the left arm but tube will not be fully filled.
C	$h_1 < \frac{P}{dg}, h_2 > 0$	R	If C_1 is opened with C_2 is closed, tube will be fully filled with liquid.
D	$h_1 + h_2 < \frac{P}{dg}, h_2 > 0$	S	If C_1 is opened with C_2 is opened, liquid will flow down continuously.
		T	After any of the above action pressure at the mid point of tube AB is non-zero

The correct match is

A) A – PQ, B – PQR, C – PT, D – ST

B) A – PQT, B – PQ, C – RST, D – RS

C) A – PQ, B – PQ, C – PRST, D – PRST

D) A – PQ, B – PQ, C – PRT, D – PRST

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CHEMISTRY **Max Marks: 60**

SECTION – I
(NUMERICAL VALUE TYPE)

This section contains **EIGHT (08)** questions.

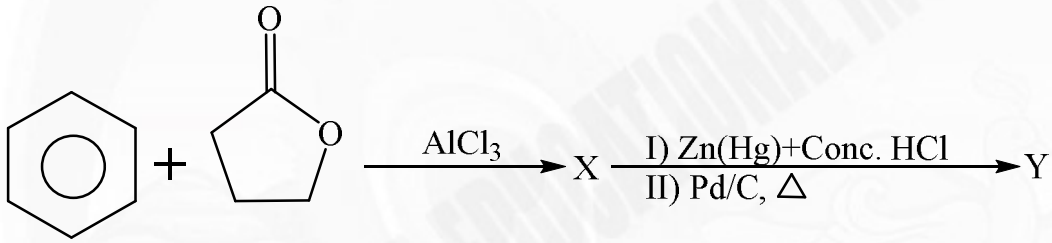
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will bee evaluated according to the following marking scheme :

Full Marks: +3 **ONLY** if the correct numerical value is entered ;
Partial Mark: 0 In all other cases.

37. $CH_3 - CCl_3 \xrightarrow{Ag,\Delta} P \xrightarrow{\text{Red hot } Fe \text{ tube}} Q$

Weight percentage of carbon in Q is_____.

38.



Weight percentage of hydrogen in the compound Y is _____.

39. The specific rotation $[\alpha]_D^{25}$ of $(S)-2-$pentanol is $+13.0$ degrees. A one molar solution of $(S)-2-$pentanol is mixed with an equal volume of one molar solution of racemic $2-$pentanol. The observed rotation, α, of the resulting solution, when measured in a polarimeter tube of path length 10cm is _____ degrees. (round off to two decimal places).

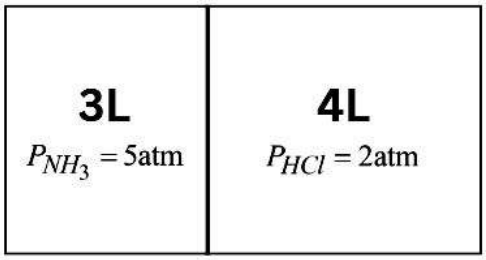
(Given: Molar mass of $2-$pentanol is $88\,g\,mol^{-1}$)

40. Calculate value of $\frac{X+Y+Z}{10}$, here X is $O-N-O$ bond angle in NO_3^- and Y is $O-N-O$ bond angle in NO_2^+ and Z is $F-Xe-F$ adjacent bond angle in XeF_4.

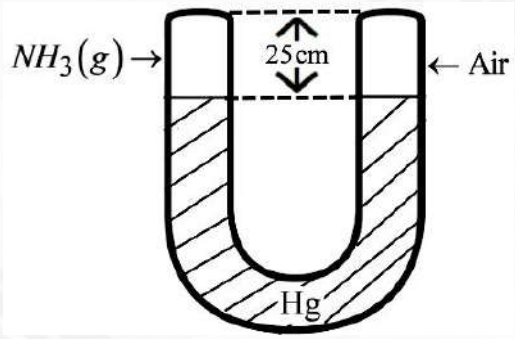
41. Considered as Covalent compound $[ML_5]$ of metals Ni and Fe have ideal square pyramidal and trigonal bipyramidal and geometries; respectively. The sum of the 90°; 120° L – M – L angles in the given structures are equal to

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42. What is the pressure (in atm) after the partition is removed, and $NH_3(g)$ and $HCl(g)$ react to form $NH_4Cl(s)$. The reaction takes place isothermally at $300K$.



43. $NH_3(g)$ is sealed in a U-tube manometer of uniform area of cross-section at a pressure of 80 cm of Hg, as shown



After sometime, due to the dissociation of NH_3, it was observed that the air column has shrunk by 5 cm. Assuming that temperature remains constant, what is the degree of dissociation of NH_3?

44. A $0.2M$ solution of $HOCl\left(pK_a = 7.5\right)$ is titrated with $0.2M$ $NaOH$. The pH after addition of $20mL$ $NaOH$ is found to be 7.50. What will be the pH of solution after $40mL$ $NaOH$ solution has beed added ?

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

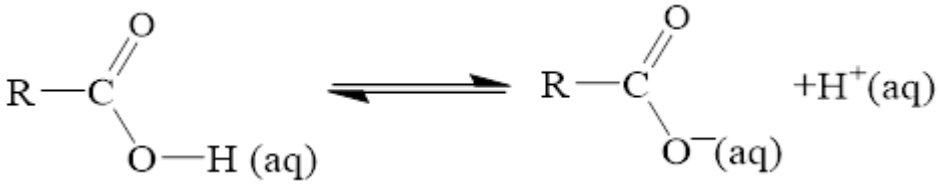
This section contains **SIX (06)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

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45. Acidic strength of weak acids can be explained by using thermodynamic data for ionization of weak acids.



Among the following, correct statement(s) is/are:

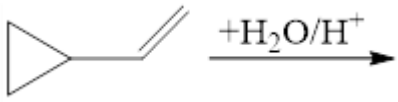
A) For the ionization of weak carboxylic acids ΔS is negative.

B) When electrons donating groups introduced in the carboxylic acid, $-T\Delta S$ become more positive

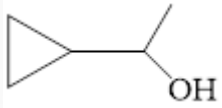
C) When electrons withdrawing groups introduced in the carboxylic acid, ΔG become less positive.

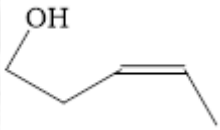
D) When electron donating group introduced in the carboxylic acid extent of hydration of carboxylate ion increases.

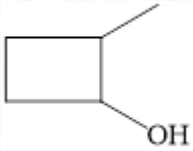
46.

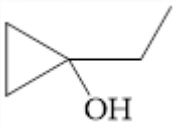


Which of the following product(s) is/are possible in the above reaction?

A) 

B) 

C) 

D) 

47. Which of the following bond length order is/are correct?

A) $H_2O_2 < F_2O_2\,(d_{O-O})$

B) $N_2O < N_2O_4 < N_2O_3\,(\text{Order of } N\text{-}N \text{ bond length})$

C) $NO_2^- < NO_3^-\,(d_{N-O})$

D) $N_2H_4 < N_2F_4\,(d_{N-N})$

48. The correct statement(s) among the following is/are

A) Energy released in the formation one $C \equiv O$ is more than the energy released when two single $C-O$ bonds are formed.

B) Energy released in the formation of one $C=O$ is less than the energy released when two single $C-O$ bonds are formed.

C) Energy released in the formation of two single $C-O$ bonds is less than the energy liberated in the formation of two single $Si-O$ bonds.

D) $Si=O$ weaker than $C=O$

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49. To the dissociation reaction $2N_2O_5(g) \rightleftarrows 4NO_2(g) + O_2(g)$,

$O_2(g)$ was added, maintaining its partial pressure constant. Which of the following statements are **INCORRECT**?

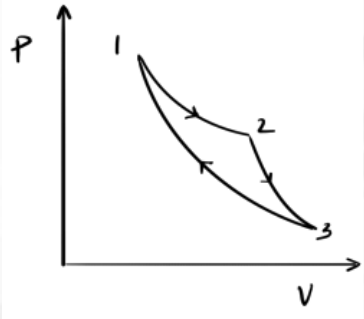
A) Kp increases and equilibrium shifts backwards.

B) Kp remains constant and the equilibrium shifts forwards.

C) Kp remains constant and the equilibrium remains undisturbed.

D) Kp remains constant and the equilibrium shifts backwards.

50. Which of the following are correct for the following process involving an ideal gas ?



$1 \rightarrow 2$ Isothermal process

$2 \rightarrow 3$ Adiabatic process

A) $\Delta U_{1-2} + \Delta U_{2-3} = W_{2-3}$

B) $\Delta S_{2\rightarrow 1} + \Delta S_{1\rightarrow 3} = \dfrac{Q_{2\rightarrow 3}}{T_2 - T_3}$

C) $W_{1-2} + W_{2-3} = W_{1-3}$

D) $Q_{1\rightarrow 3} + W_{1\rightarrow 3} + W_{3\rightarrow 2} = 0$

SECTION – III
(MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

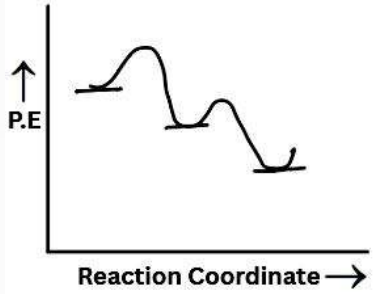
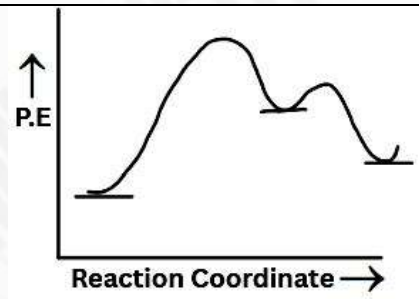
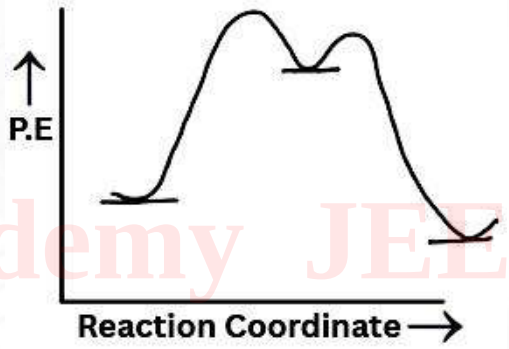
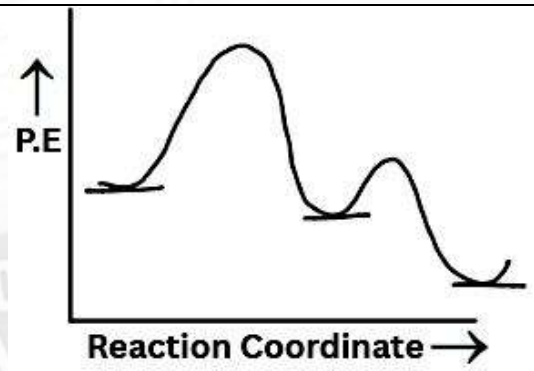
Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosed (i.e. the question is unanswered);
Negative Marks: –1 In all other cases.

51. Halogen of ethane in the presence of sunlight follows free radical substitution mechanism. It is chain reaction. Two steps in the chain propagation is given below

$$CH_3 - CH_2 - H + X^{\cdot} \longrightarrow CH_3 - \dot{C}H_2 + HX$$

$$CH_3 - \dot{C}H_2 + X - X \longrightarrow CH_3 - CH_2 - X + X^{\cdot}$$

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List – I (Halogenation)		List – II (Potential energy diagram for chain propagation)	
P)	Fluorination	**I)**	
Q)	Chlorination	**II)**	
R)	Bromination	**III)**	
S)	Iodination	**IV)**	

The correct match is

A) P-IV; Q-I; R-II; S-III **B)** P-I; Q-III; R-IV; S-II

C) P-I; Q-IV; R-II; S-III **D)** P-I; Q-IV;R-III; S-II

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52. Match List – I with List – II

List - I		List - II	
P	Boiling point (K)	I	$B > Al > Ga > In > Tl$
Q	Ionic radius	II	$Sb^{3+} < Bi^{3+} < N^{3-} < P^{3-} < As^{3-}$
R	Atomic radius (pm)	III	$Tl > In > Al > Ga > B$
S	Density $\left(g / cm^3\right)$ at 298 K	IV	$N < P < As < Sb < Bi < Pb$

The correct match is

A) P – I; Q – II; R – II; S – II **B)** P – IV; Q – II; R – III; S – IV

C) P – I; Q – II; R – III; S – IV **D)** P – IV; Q – II; R – II; S – I

53. Match the **Columns**

COLUMN – I describes the orbital for a single electron species and **COLUMN – II** gives the geometrical description of intersection between a pair of nodal surfaces for the given orbital

COLUMN – I		COLUMN – II	
P)	$3d_{xy}$	**I)**	Point
Q)	$4d_{x^2-y^2}$	**II)**	Circle
R)	$3d_{z^2}$	**III)**	Straight line
S)	$4d_{z^2}$	**IV)**	Ellipse

A) P – II; Q – I, III; R – IV; S – I, IV **B)** P – I; Q – I, II; R – IV; S – I, II

C) P – III; Q – II, III; R – I; S – I, II **D)** P – IV; Q – II, IV; R – III; S – I, III

54. Match **COLUMN – I** and **COLUMN – II** for processes involving an ideal monoatomic gas $\left(C_v, m = \frac{3R}{2}\right)$

W = work done on the gas; Q = heat supplied to the gas.

COLUMN – I		COLUMN – II	
1)	Reversible process with $Q = W$	**P)**	$P_1V_1 = P_2V_2$
2)	Reversible process with $Q = -W$	**Q)**	$T_1V_1^{2/3} = T_2V_2^{2/3}$
3)	Reversible process with $Q = 0$	**R)**	$T_1V_1^{4/3} = T_2V_2^{4/3}$
4)	Free expansion with $Q = 0$	**S)**	$P_1V_1^{7/3} = P_2V_2^{7/3}$

A) 1-S; 2-P; 3-Q; 4-R **B)** 1-R; 2-P; 3-S; 4-P

C) 1-R; 2-P; 3-Q; 4-P **D)** 1-S; 2-Q; 3-Q; 4-P

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